LECTURE 6

DATA ANALYSIS

ANALYZING DATA

LECTURE CONTENT

- Logic functions
- Pivot tables
- Sumif, averageif, countif, etc.
- Sumifs, maxifs, minifs, averageifs, etc.
- VLookUP

IF() FUNCTION

IF() function returns one value if a logical expression is 'TRUE' and another if it is 'FALSE'.

Syntax

```
=IF(logical_test, [value_if_true], [value_if_false])
```

- `logical_test` : criteria you want to test
- `value_if_true` : value that you want returned if the result is TRUE
- `value_if_false` : value that you want returned if the result is FALSE

Example

	A	B	C	D
1	**Name**	**Test result**	**Pass OR Fail**	
2	Aidana	85	pass	
3	Dina	67	pass	
4	Nurbek	81	=IF(B4>60, "pass", "fail")	
5	Kamal	70	pass	
6	Laura	92	pass	
7	Rauan	95	pass	
8	Kerim	50	fail	
9	Meyrambek	61	pass	
10	Bekasyl	45	fail	

AND() FUNCTION

The **AND** function returns true if all of the provided arguments are logically true, and false if any of the provided arguments are logically false. The AND function can be used as the logical test inside the IF function to avoid extra nested IFs, and can be combined with the OR function.

Syntax

```
=AND(logical_test1, logical_test2, ...)
```

- `TRUE` if **all** arguments evaluate to `TRUE`
- `FALSE` if **one or more** arguments evaluate to `FALSE`

Formula	Output
=AND(TRUE,TRUE)	TRUE
=AND(TRUE,FALSE)	FALSE
=AND(FALSE,FALSE)	FALSE
=AND(FALSE,TRUE)	FALSE

Example

Test if a number in A24 is greater than zero and less than 100,
use =AND(A24>0,A24<100).

	A	B	C
23	VALUE	RESULT	
24	5	=AND(A24>0, A24<100)	
25	231	FALSE	
26	76	TRUE	
27	-40	FALSE	
28	61	TRUE	
29	120	FALSE	

OR() FUNCTION

The **OR** function returns true if any of the provided arguments are logically true, and false if all of the provided arguments are logically false.

Syntax

```
=OR(logical_test1, logical_test2, ...)
```

- `TRUE` if **any** of the arguments evaluate to `TRUE`
- `FALSE` if **all** of the arguments evaluate to `FALSE`

Formula	Description	Result
=OR(TRUE,TRUE)	All arguments are TRUE	TRUE
=OR(TRUE,FALSE)	One argument is FALSE	TRUE
=OR(1=1,2=2,3=3)	All arguments are TRUE	TRUE
=OR(1=2,2=3,3=4)	All arguments are FALSE	FALSE

Example

Test if a number in A24 is greater than zero or less than 100,
use =OR(A24>0,A24<100).

	A	B	C
23	**VALUE**	**RESULT**	
24	5	=OR(A24>0, A24<100)	
25	231	TRUE	
26	76	TRUE	
27	-40	TRUE	
28	61	TRUE	
29	120	TRUE	

NESTED IF(), AND(), OR() FUNCTIONS

Suppose you have a list of students with their marks and attendance. Only those students get the scholarship who have a score of greater than 80 and attendance of more than 80%. You can quickly identify such students using the AND condition within the IF function.

Example

	A	B	C	D	E	F
13	**Name**	**Score**	**Attendance**	**Scholarship**		
14	Aidana	85	81%	=if(AND(B14>=80,C14>=80%),"Yes", "No")		
15	Dina	67	85%	No		
16	Nurbek	81	90%	Yes		
17	Kamal	70	65%	No		
18	Laura	92	90%	Yes		
19	Rauan	95	85%	Yes		
20	Kerim	50	70%	No		
21	Meyrambek	61	50%	No		
22	Bekasyl	45	65%	No		

MULTIPLE CONDITIONS WITH IF() FUNCTION

Task: assign a grade to a student based on the marks obtained.
For example, a student gets "A+" grade if he/she gets between 95-100, "A-" for marks between 90 and 94, "B+" for marks between 85 and 89, etc. (as shown below):

Example

	A	B	C	D	E	F	G	H
30	**Score**	**Grade**						
31	9	=IF(and(A31>=95,A31<=100), "A+",if(and(A31>=90,A31<=94),"A-",if(and(A31>=85,A31<=89),"B+")))						
32	90	A-						
33	87	B+						

SUMIF
AVERAGEIF

SUMIF - returns a conditional sum across a range.

	A	B	C	D	E
1	**Expense**	**Today**			
2	Coffee	$4.00			
3	Newspaper	$1.00			
4	Taxi	$10.00			
5	Golf	$26.00			
6	Taxi	$8.00			
7	Coffee	$3.50			
8	Gas	$46.00			
9	Restaurant	$31.00			
10					
11					
12	**range**	**criteria**	**sum_range**	**Sum**	**Formula**
13	A2:A9	Taxi	B2:B9	$18.00	=SUMIF(A2:A9, "Taxi", B2:B9)

AVERAGEIF - returns the average of a range depending on criteria.

	A	B	C	D	E
1	**Expense**	**Today**			
2	Coffee	$4.00			
3	Newspaper	$1.00			
4	Taxi	$10.00			
5	Golf	$26.00	**Average**	**Formula**	
6	Taxi	$8.00	$3.75	=AVERAGEIF(A2:A9, "Coffee", B2:B9)	
7	Coffee	$3.50			
8	Gas	$46.00			
9	Restaurant	$31.00			
10					

COUNTIF

COUNTIF

Returns a conditional count across a range.

	A	B	C	D
1	**Expense**	**Today**		
2	Coffee	$4.00		
3	Newspaper	$1.00		
4	Taxi	$10.00		
5	Golf	$26.00		
6	Taxi	$8.00		
7	Coffee	$3.50		
8	Gas	$46.00		
9	Restaurant	$31.00		
10	...	...		
11				
12				
13	**range**	**criteria**	**Result**	**Formula**
14	A2:A9	"Taxi"	2	=COUNTIF(A2:A9, "Taxi")
15				

SUMIFS

The only difference between SUMIFS & SUMIF functions is that SUMIFs can check for multiple criteria at once, while SUMIF can check for one criterion at a time.

	A	B	C	D	E
34	**Quantity**	**Product**	**Salesperson**		
35	5	T-shirt	Tom		
36	2	T-shirt	Saule		
37	15	Dress	Saule		
38	3	Gloves	Tom		
39	22	Sweater	Saule		
40	12	Gloves	Tom		
41	10	Sweater	Tom		
42	1	T-shirt	Tom		
43					
44	Salesperson	Product	Product sold		
45	Tom	T-shirt	? =SUMIFS(A35:A42,B35:B42,"T-shirt",C35:C42,"Tom")		
46					

44	**Salesperson**	**Product**	**Product sold**	
45	Tom	T-shirt	6	=SUMIFS(A35:A42,B35:B42,"T-shirt",C35:C42,"Tom")
46			31	=SUMIF(C35:C42,"Tom", A35:A42)
47				

MAXIFS

The MAXIFS function returns the largest numeric value that meets one or more criteria in a range of values

	A	B	C	D	E	F
34	**Quantity**	**Product**	**Salesperson**			
35	5	T-shirt	Tom			
36	2	T-shirt	Saule			
37	15	Dress	Saule			
38	3	Gloves	Tom			
39	22	Sweater	Saule			
40	12	Gloves	Tom			
41	10	Sweater	Tom			
42	1	T-shirt	Tom			
43						
44	**Salesperson**	**Product**	**Product sold**			
45	Tom	T-shirt	5	=MAXIFS(A35:A42,B35:B42,"T-shirt",C35:C42,"Tom")		

MINIFS

	A	B	C	D	E	F
34	**Quantity**	**Product**	**Salesperson**			
35	5	T-shirt	Tom			
36	2	T-shirt	Saule			
37	15	Dress	Saule			
38	3	Gloves	Tom			
39	22	Sweater	Saule			
40	12	Gloves	Tom			
41	10	Sweater	Tom			
42	1	T-shirt	Tom			
43						
44	**Salesperson**	**Product**	**Product sold**			
45	Tom	T-shirt	1	=MINIFS(A35:A42,B35:B42,"T-shirt",C35:C42,"Tom")		

The MINIFS function returns the minimum value among cells specified by a given set of conditions or criteria.

AVERAGEIFS

The AVERAGEIFS function calculates the average of the numbers in a range that meet supplied criteria

	A	B	C	D	E	F	G
34	Quantity	Product	Salesperson				
35	5	T-shirt	Tom				
36	2	T-shirt	Saule				
37	15	Dress	Saule				
38	3	Gloves	Tom				
39	22	Sweater	Saule				
40	12	Gloves	Tom				
41	10	Sweater	Tom				
42	1	T-shirt	Tom				
43							
44	Salesperson	Product	Product sold				
45	Tom	T-shirt	3	=AVERAGEIFS(A35:A42,B35:B42,"T-shirt",C35:C42,"Tom")			

COUNTIFS

	A	B	C	D	E	F
34	Quantity	Product	Salesperson			
35	5	T-shirt	Tom			
36	2	T-shirt	Saule			
37	15	Dress	Saule			
38	3	Gloves	Tom			
39	22	Sweater	Saule			
40	12	Gloves	Tom			
41	10	Sweater	Tom			
42	1	T-shirt	Tom			
43						
44	Salesperson	Product	Product sold			
45	Tom	T-shirt	2	=COUNTIFS(B35:B42,"T-shirt",C35:C42,"Tom")		

The COUNTIFS function applies criteria to cells across multiple ranges and counts the number of times all criteria are met.

PIVOT TABLES

A PivotTable is a powerful tool to calculate, summarize, and analyze data that lets you see comparisons, patterns, and trends in your data.

The purpose of pivot tables is to offer user-friendly ways to quickly summarize large amounts of data. They can be used to better understand, display, and analyze numerical data in detail — and can help identify and answer unanticipated questions surrounding it.

	A	B	C	D	E
1	division	subdivision	product numbe	number of units	price per unit
2	east	1	$1	14	$10
3	east	2	$1	15	$11
4	west	1	$1	11	$10
5	west	2	$1	21	$9
6	east	3	$1	16	$8
7	west	3	$1	18	$12
8	east	4	$1	11	$9
9	east	1	$2	10	$9
10	east	2	$2	9	$13
11	west	1	$2	12	$10
12	west	2	$2	15	$10
13	east	3	$2	12	$9
14	west	3	2	16	$12
15	east	4	2	12	$9

	A	B	C	D
1	*division*	*subdivision*	SUM of number of units	AVERAGE of price per unit
2	east	1	24	$9.50
3		2	24	$12.00
4		3	28	$8.50
5		4	23	$9.00
6	east Total		99	$9.75
7	west	1	23	$10.00
8		2	36	$9.50
9		3	34	$12.00
10	west Total		93	$10.50
11	**Grand Total**		**192**	**$10.07**

VLOOKUP

VLOOKUP stands for Vertical Lookup. As the name specifies, VLOOKUP helps you look for a specified value by searching for it vertically across the sheet.

	A	B	C
1	**Student ID**	**Grade**	
2	N444	90	
3	N333	100	
4	N222	85	
5	N111	80	
6			
7			
8	**Student ID**	**Grade**	**Formula**
9	N111	80	=VLOOKUP(A9,$A$2:$B$5, 2, FALSE)
10	N222	85	=VLOOKUP(A10,$A$2:$B$5, 2, FALSE)
11	N333	100	=VLOOKUP(A11,$A$2:$B$5, 2, FALSE)
12	N444	90	=VLOOKUP(A12,$A$2:$B$5, 2, FALSE)

In this example, VLOOKUP searches down the first column for a student ID and returns the corresponding grade